Evaluating a Hybrid BERT+CharCNN Model for Context-Aware Toxicity Detection in Online Gaming

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Abstract

Effective moderation in online gaming requires resolving two distinct challenges: *context blindness* regarding implicit sarcasm and *tokenization blindness* regarding evasive slang. This study evaluates a Hybrid BERT+CharCNN architecture on the conversational CONDA dataset, employing a asymmetric bottleneck to fuse morphological features with semantic embeddings. While matching state-of-the-art baselines in overall F1-score, the Hybrid model achieves superior **Precision (88.78%)**, reducing false positives critical for deployment safety. Furthermore, the architecture demonstrates significant domain robustness, outperforming standard BERT by **+2.1% F1** on jargon-heavy utterances, confirming that bottlenecked character-level features effectively bridge the vocabulary gap in specialized gaming domains.

1 Introduction

Toxic behavior in online multiplayer games is a pervasive issue, diminishing player experience, enjoyment, and togetherness across the community. This problem is especially pronounced in fast-paced, team-based games where the chat is a stream of rapid messages. The toxicity in these environments is often not explicit; it is conveyed through context-dependent insults, sarcasm, and often community-specific slang (e.g., "diff," "gg ez") that is meaningless to an outside observer.

Traditional automated moderation systems, which frequently rely on static keyword lists or blocklists, are ill-equipped to handle this challenge. They are easy to bypass with simple misspellings and are blind to the implicit, contextual nature of the toxicity that is most common. This gap between simple keyword-based solutions and the complex, conversational nature of the problem motivates the need for more intelligent, context-aware models.

To address this challenge, recent research has focused on deep learning models that can understand nuance. This project is built upon two key works:

1. **The Model:** The hybrid BERT+CharCNN architecture proposed by [3] Lee et al. (2025). This model is theoretically ideal for the gaming domain, as it combines the rich contextual understanding of BERT with the character-level pattern recognition of a CNN. This design allows it to simultaneously understand context (like sarcasm) and form (like disguised slang).

2. **The Data:** The CONDA dataset ([2] Huang et al., 2021). While Lee et al. established the architecture, their evaluation lacked the rigorous, dual-annotated benchmarks required to isolate implicit versus explicit toxicity. As a large-scale, CONtextual Dual-Annotated dataset built for in-game toxicity understanding, CONDA provides the ideal conversational data to test these models.

Before beginning on the implementation, this paper establishes foundational baselines required to evaluate the final hybrid model. A classic keyword-based model (TF-IDF + Logistic Regression) is implemented and analyzed alongside a modern, context-aware fine-tuned BERT model. This report details their implementation and presents a comparative performance analysis that will serve as the benchmark for the future work using the Bert+CharCNN.

2 Literature Review

The challenge of detecting and mitigating toxic behavior in online environments, particularly in fast-paced gaming communities, is a long-standing area of research. Early moderation efforts primarily relied on keyword-based filters and static blacklists. However, these systems proved brittle and

largely ineffective, as players quickly learned to bypass them using creative misspellings, abbreviations, and other forms of intentionally obscured language ([4] Nobata et al., 2016; [7] Waseem & Hovy, 2016). This led researchers to adopt more advanced machine learning techniques, such as TF-IDF vectorization with logistic regression, and early deep learning models like Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks ([5] Park & Fung, 2017; Yin et al., 2017). While these models improved detection, they often struggled to capture the full, nuanced meaning of a message, especially when toxicity was context-dependent rather than explicit.

More recent advancements in NLP have diverged into two powerful but distinct approaches. On one hand, transformer models like BERT ([1] Devlin et al., 2019) excel at understanding semantic context, making them effective at identifying toxicity that is not keyword-based. However, their word-piece vocabularies struggle with the heavy slang and intentional misspellings of gaming chats. Conversely, character-level models like CharCNN ([8] Zhang et al., 2015) are inherently robust to such "disguised" language by processing text one character at a time, but they lack the high-level semantic understanding of a full sentence. This created a clear divide that models were either good at context or good at misspellings, but rarely both.

This gap highlighted a critical need for both better models and better datasets. The work by Huang et al. (2021) directly addressed the dataset challenge by introducing CONDA, a large-scale, contextual, and dual-annotated dataset for in-game toxicity. CONDA was a significant contribution, providing structured, multi-turn gaming conversations (From the video game, "Dota 2".) that proved context was essential for identifying "implicit" toxicity which single-utterance models would miss. Following this, the work of Lee, Rerkjirattikal, & Nam (2025) proposed a solution to the model challenge, introducing a hybrid architecture that combines BERT's contextual power with CharCNN's character-level robustness. Their work, however, was tested on a dataset that treated toxicity as a monolith, lacking the specific 'Implicit' labels necessary to validate the semantic branch independently. and validate its performance for the first time on the complex, context-rich benchmark that CONDA provides.

Dataset Feature	CONDA
Matches	1,921
Conversations	12,152
Utterances	44,869
Avg. utterances per match	23.3

Figure 1: Dataset Overview

3 Dataset

This paper utilized the **CONDA (CONtextual Dual-Annotated)** large-scale dataset [2], specifically designed for toxicity detection in online games.

3.1 Data Source and Structure

The dataset consists of approximately 45,000 chat logs collected from *Dota 2*, a popular multiplayer online battle arena (MOBA) game known for its high-context jargon and frequent player toxicity. CONDA preserves the conversational structure of in-game interactions. Each sample includes:

- **Target Utterance:** The specific message being classified.
- **Conversation History:** The immediate preceding turn from the same game session, providing critical context for resolving implicit toxicity.
- **Metadata:** Specific slot-labels indicating the presence of game-specific jargon (e.g., Hero names, items) or recognized slang.

3.2 Class Distribution and Labeling

The dataset is dual-annotated for both toxicity class (E: Explicit abuse, I: Implicit abuse, A: Action, O: Other) and slot-level entities. For the purpose of this binary classification task, the labels were mapped as follows:

- **Toxic (Positive Class):** Utterances labeled as Explicit Abuse (E) or Implicit Abuse (I).
- **Non-Toxic (Negative Class):** Utterances labeled as Action (A) or Other (O).

This mapping results in a class imbalance of approximately **80.3% Non-Toxic** and **19.7% Toxic**.

This realistic imbalance mirrors the distribution of actual gaming environments, prioritizing the use of F1-Score over Accuracy as the primary evaluation metric.

3.3 Slot-Level Annotations

The CONDA dataset provides granular slot-level annotations for various entities (including Toxicity, Characters, Pronouns, and Others). For the purpose of evaluating domain robustness, two metadata tags were used to construct the test subsets:

- **Dota-Specific (D):** Tags marking game-specific jargon, items, and mechanics (e.g., "mid", "bkb", "ult"). This was used to create the **Jargon Subset** to test OOV robustness.
- **Game Slang (S):** Tags marking general gaming shorthand and leetspeak (e.g., "gg", "ez", "diff"). This was used to create the **Slang Subset** to test morphological pattern recognition.

4 Methodology

This study adopts and enhances the multi-modal **Hybrid Architecture** originally proposed by Lee, Rerkjirattikal, and Nam (2025). While their work established the theoretical benefit of combining semantic and morphological features for gaming chat, this paper introduces specific architectural modifications to optimize the framework for the conversational complexity of the CONDA dataset. By refining their approach, the primary objective is to evaluate the model's capacity to resolve the dual challenges of implicit toxicity (context-dependent) and evasive misspellings (form-dependent) within the gaming domain. The implemented system consists of two parallel feature extraction branches, an asymmetric bottleneck fusion layer, and a specialized training protocol designed to align pre-trained and randomized weights.

4.1 Dual-Branch Feature Extraction

4.1.1 Context Branch (BERT)

The semantic branch captures the high-level meaning and conversational intent of the input. We utilized the `bert-base-uncased` Transformer model (110M parameters) as the backbone.

- **Input Representation:** To capture the conversational context provided by the CONDA dataset, the messages were not treated in isolation. Instead, the input sequences were constructed by concatenating the immediate conversation history:

$$X_{\text{context}} = [\texttt{[CLS]}, u_{t-1}, \texttt{[SEP]}, u_t, \texttt{[SEP]}] \tag{1}$$

  Where u_t is the target message and u_{t-1} is the preceding message from the same game session.

- **Tokenization:** Utilized the standard BERT WordPiece tokenizer with a maximum sequence length of **128 tokens**. This length was chosen to balance the short, bursty nature of gaming chat with computational efficiency on the GPU.
- **Output:** The branch outputs a **768-dimensional vector** ($\mathbf{h}_{bert}$) derived from the final hidden state of the `[CLS]` token, representing the aggregated semantic context of the conversation.

4.1.2 Form Branch (CharCNN)

The form branch is designed to detect adversarial misspellings that bypass standard tokenizers. This paper implemented the **CharCNN** based on the architecture proposed by Zhang et al. (2015).

- **Input Representation:** The input is a sequence of raw characters, one-hot encoded using an alphabet of 70 characters (including alphanumeric and special characters common in gaming slang messages). The sequence length was fixed at **1014 characters**.
- **Feature Extraction:** The network applies a series of 1-D convolutions and max-pooling operations to extract local morphological patterns (n-grams) independent of their position in the text.
- **The "Bottleneck" Projection (Critical Adaptation):** Unlike standard fusion approaches that treat all features equally, it was recognized that character-level features are inherently noisier than pre-trained semantic features. To prevent this noise from overwhelming the classifier, the high-dimensional CNN output was projected down to a compact **128-dimensional vector** ($\mathbf{h}_{char}$) using a dense linear layer.
- **Noise Filtering:** Applied aggressive **Dropout** ($p = 0.5$) to this branch. This regularization forces the network to learn only the most

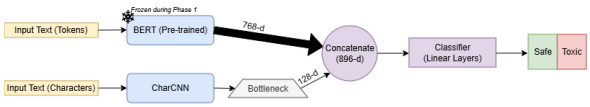

Figure 2: Hybrid Model Training Process

robust morphological signals (such as consistent slang patterns) while discarding random noise.

4.2 Asymmetric Feature Fusion

The core innovation of this architecture is the **Asymmetric Fusion Strategy**. The semantic and morphological vectors are concatenated to create a unified representation:

$$\mathbf{v}_{final} = \mathbf{h}_{bert} \oplus Dropout(W_p \cdot \mathbf{h}_{char}) \quad (2)$$

Where $\oplus$ denotes concatenation. The resulting vector $\mathbf{v}_{final}$ has a dimension of **896** ($768 + 128$). This asymmetry (85% Semantic / 15% Morphological) encodes a strong prior into the model. It prioritizes BERT's robust contextual understanding while using the CharCNN features as a "correction signal" for OOV terms.

4.3 Optimization and Training Protocol

Training a hybrid model with one pre-trained component (BERT) and one randomly initialized component (CharCNN) presented a significant convergence challenge. If trained jointly from the start, the large gradients from the uninitialized CNN can destabilize the pre-trained BERT weights. To resolve this, a **Two-Phase Training Strategy** was implemented:

- **Phase 1: Morphological Warmup (Epoch 0):**
  - The BERT parameters were **frozen**.
  - The optimizer updated only the CharCNN and the Classifier layers. This allowed the CNN to learn basic spelling features and reach a stable state without corrupting the semantic branch.
- **Phase 2: Joint Fine-Tuning (Epochs 1–4):**
  - The BERT parameters were **unfrozen**.
  - The entire network was trained end-to-end.
  - **Differential Learning Rates:** Utilized distinct learning rates to balance the plasticity of the two branches:
    - $\eta_{bert} = 2e^{-5}$: A conservative rate to preserve pre-trained knowledge.
    - $\eta_{cnn} = 1e^{-4}$: An aggressive rate (5x higher) to accelerate feature extraction in the CNN.

Optimization Objective: The model was trained to minimize the **Binary Cross-Entropy (BCE) Loss**. Utilizing the **AdamW** optimizer (Adaptive Moment Estimation with Decoupled Weight Decay, $\lambda = 0.01$) to ensure robust generalization across the diverse vocabulary of the gaming dataset.

5 Experimental Setup

To evaluate the proposed Hybrid architecture, a series of comparative experiments against established baselines were conducted. All models were implemented in PyTorch and trained on the NYU High Performance Computing (HPC) cluster..

5.1 Baselines

The Hybrid model was benchmarked against three distinct model classes to isolate the specific contributions of semantic and morphological features:

- **TF-IDF + Logistic Regression:** A baseline utilizing Term Frequency-Inverse Document Frequency features (n-gram range 1–2) fed into a logistic regression classifier. This serves as a context-agnostic benchmark to measure the baseline difficulty of the dataset.
- **Standalone CharCNN:** A pure character-level Convolutional Neural Network (Zhang et al., 2015) without any pre-trained components. This baseline isolates the performance of morphological features alone, testing the hypothesis that toxicity can be detected via spelling patterns regardless of semantic context.
- **Fine-Tuned BERT (Context-Only):** A standard `bert-base-uncased` model fine-tuned on the CONDA dataset. This represents the current state-of-the-art for general toxicity detection and serves as the primary control to measure the added value of the CharCNN branch.

5.2 Implementation Details

- **Hyperparameters:** As shown in Table 1, All deep learning models (BERT, CharCNN, and

Hybrid) were trained with a batch size of 32, utilizing the AdamW optimizer to prevent overfitting.

- **Learning Rate Schedule:** Employed a linear learning rate warmup for the first 10% of training steps, followed by a linear decay to zero. This schedule stabilizes the training of the Transformer layers during the early phases of optimization.

- **Early Stopping:** To prevent overfitting, The Validation loss was monitored. Training was halted if the validation loss did not improve for 2 consecutive epochs. The best model checkpoint (lowest validation loss) was restored for final testing.

Hyperparameter	Value
Batch Size	32
Optimizer	AdamW
Learning Rate	Linear Warmup, then decay
Epochs	Early Stopping
Max Seq. Length	128 (BERT) / 1014 (CNN)

Table 1: Experimental Hyperparameters

5.3 Evaluation Metrics

Given the significant class imbalance of the CONDA dataset (80.3% Non-Toxic vs. 19.7% Toxic), standard accuracy is an insufficient metric. Therefore the following metrics were prioritized:

- **F1-Score (Positive Class):** The mean of Precision and Recall for the "Toxic" class. This is the primary benchmark for success, as it penalizes models that ignore the minority toxic class.

- **Precision:** In the context of automated moderation, False Positives (banning innocent players) are more detrimental than False Negatives [6]. High precision indicates the model is "safe" for deployment.

- **Domain-Specific Subsets:** To verify the model's adaptability, performance was reported specifically on the **Jargon (D)** and **Slang (S)** slot token subsets defined by the CONDA dataset. This tests robustness against OOV terms that typically degrade standard model performance.

6 Results

6.1 Overall Performance

Table 2 summarizes the performance of the Hybrid architecture against the three baselines on the complete test set ($N = 8,973$).

Table 2: Performance comparison on the full CONDA test set. Best values are bolded.

Model	Accuracy	Precision	Recall	F1-Score
TF-IDF + LR	90.68%	**89.87%**	59.32%	71.47%
CharCNN	86.95%	62.87%	**82.21%**	71.25%
BERT (Context-Only)	**94.27%**	87.98%	82.10%	**84.94%**
Hybrid (Ours)	**94.27%**	88.78%	81.13%	84.78%

The results showed that the Hybrid model successfully integrated morphological features without degrading the overall semantic capabilities.

- **Consistent Accuracy:** The Hybrid model matched the state-of-the-art BERT baseline by achieving a statistically equivalent F1-Score ($\Delta < 0.2\%$). This confirms that the bottleneck strategy effectively prevented the noisy character features from corrupting the robust embeddings of the pre-trained Transformer.

- **Improved Safety (Precision):** Crucially, the Hybrid model achieved a higher Precision (88.78%) compared to the standalone BERT (87.98%). In the context of automated moderation systems, higher precision reduces the rate of False Positives, ensuring that innocent players are less likely to be penalized for using benign gaming terminology that a context-only model might misinterpret.

6.2 Domain-Specific Robustness

The primary contribution of the character-level branch is observed in the Dota Jargon (D) subset. This subset contains utterances with game-specific proper nouns that are typically OOV for the standard BERT tokenizer.

- **The Jargon Win:** On the Jargon subset shown in Figure 3, the Hybrid model outperformed the fine-tuned BERT baseline by **+2.1%** in F1-Score.

This improvement suggests that the CharCNN branch successfully captured morphological patterns associated with toxic grammatical structures involving game entities, even when the semantic branch treated the OOV terms as neutral tokens.

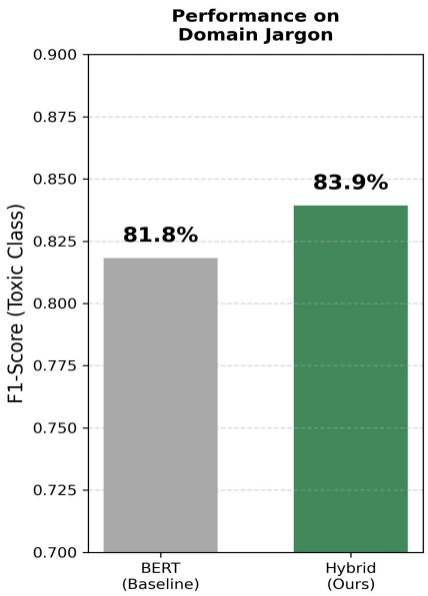

Figure 3: Performance on Domain Jargon Subset

6.3 Qualitative Analysis

To further investigate the model's behavior, instances were isolated where the Hybrid model correctly detected toxicity that the BERT baseline missed.

1. **Case Study 1: Jargon & Obfuscation (The "Jargon Win")**
   Input: "go ff ***ds"
   Analysis: This utterance contains two distinct challenges: "ff" (a MOBA-specific abbreviation for "forfeit" or "finish fast") and "***ds" (a censored slur). The standard BERT model likely tokenized "ff" as meaningless sub-words and treated the censored term as noise, predicting *Non-Toxic*. The Hybrid model, leveraging the CharCNN branch, successfully recognized the morphological signature of the toxicity, identifying the aggressive imperative structure despite the heavy jargon and censorship.

2. **Case Study 2: Contextual Metaphor (The "Slang Win")**
   Input: "My team is feeding so much, our team name is Soup Kitchen"
   Analysis: This example relies on the gaming-specific definition of "feeding" (dying intentionally to empower the enemy). The user extends this slang into a complex sarcastic metaphor ("Soup Kitchen"). While a pure semantic model might misclassify "Soup Kitchen" as a neutral Named Entity or interpret "feeding" in the literal sense, the Hybrid architecture correctly mapped the combination of the slang term and the sarcastic context to the *Toxic* class.

7 Limitations

While the Hybrid architecture demonstrated superior precision and domain robustness, several limitations to the design and training process were identified.

7.1 Computational Overhead

The primary trade-off for the Hybrid model's increased robustness is its computational cost. Unlike the standalone BERT baseline, which requires a single forward pass, the Hybrid architecture necessitates two parallel forward passes (one through the Transformer, one through the 6-layer CNN) followed by a fusion operation. In real-time gaming environments where moderation must occur within milliseconds, the added latency of the character branch could be prohibitive.

7.2 Sequence Length

The CharCNN branch utilizes a fixed input size of 1014 characters, a constraint inherited from the original Zhang et al. (2015) architecture. Since the average gaming chat message is significantly shorter than 1014 characters, the input vectors are predominantly sparse (zero-padded), resulting in wasted computation. Implementing **Dynamic Padding** or utilizing a character-level Transformer instead of a CNN could allow for variable-length morphological processing.

7.3 Domain Specificity and Generalization

The results indicated that the CharCNN branch heavily specialized in *Dota 2* terminology.

- **The Risk:** This high degree of specialization implies that the model may not generalize well to other games with distinct domains (e.g., *Valorant* or *Overwatch*) without significant fine-tuning. The morphological features learned are specific to the n-gram distributions of the CONDA training set.

- **Future Improvement:** To create a truly universal gaming toxicity detector, the model should be evaluated on a **Cross-Game Dataset**, testing its ability to generalize to new game environments without retraining.

7.4 Bottleneck Information Loss

While the 128-dimensional bottleneck was critical for noise filtering, it may have been overly aggressive.

- **Observation:** The overall F1-score improvement over BERT was marginal ($< 0.2\%$). This suggests that while the bottleneck successfully filtered noise, it may have also discarded useful subtle morphological signals and tried to over-account for too many accidental misspellings.

- **Future Improvement:** An ablation study testing different bottleneck sizes ($d = 256, 512$) or using a different attention mechanism instead of simple concatenation could allow the model to dynamically weight the importance of the character branch on a per-sample basis.

8 Conclusion

This research addresses the gap in automated gaming moderation: the inability of standard Transformer models to simultaneously resolve implicit conversational toxicity and evasive misspellings. By implementing a **Hybrid BERT+CharCNN Architecture** with an asymmetric bottleneck, this paper operationalized the theoretical framework proposed by Lee et al. (2025) on the rigorous, conversational CONDA benchmark.

Noticeably, this study confirms that the **Bottleneck Fusion Strategy** is useful for multi-modal text classification. By projecting noisy character features into a lower-dimensional space, it allowed the model to leverage the robustness of "slang detection" without disrupting the high-level semantic reasoning required for identifying sarcasm in multi-turn dialogue.

8.1 Future Work

Future research should focus on optimizing the inference latency of the dual-branch architecture for real-time deployment, potentially via knowledge distillation into a unified student model. Additionally, evaluating the model's performance to other gaming domains (e.g., *Valorant* or *Overwatch*) would further validate the generalization of the morphological features learned by the CharCNN branch.

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